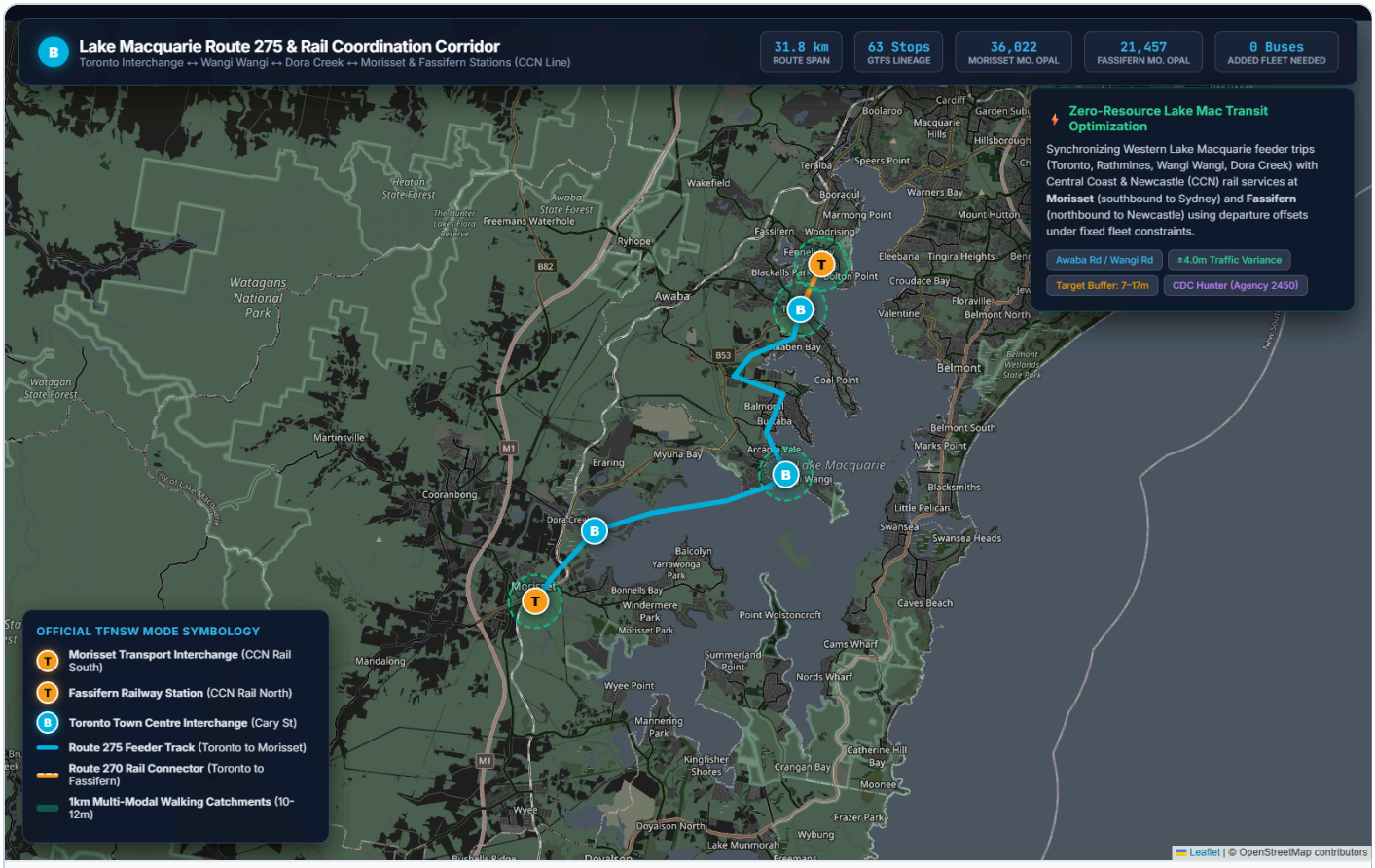
Lake Macquarie Route 275: Zero-Resource Rail Coordination

Connecting Western Lake Mac Communities to Morisset & Fassifern Rail Gateways



Interactive Corridor Map: Route 275 feeder track (blue) connecting Toronto to Morisset Interchange; Route 270 rail link (orange dashed) connecting to Fassifern Station; Orange [T] roundels: CCN rail hubs; Green rings: 1km pedestrian walking catchments.

31.8 km	63 Stops	36,022	21,457	0 Buses	7–17m
CORRIDOR LENGTH	GTFS GROUND TRUTH	MORISSET MO. OPAL	FASSIFERN MO. OPAL	EXTRA FLEET NEEDED	TARGET TRANSFER BUFFER

The Everyday Commuter Problem

Route 275 serves **Toronto, Carey Bay, Kilaben Bay, Rathmines, Fishing Point, Wangi Wangi, Arcadia Vale, and Dora Creek**.

Under existing timetables, feeder buses frequently arrive at Morisset or Toronto **3 to 5 minutes after the hourly train has already departed**, stranding commuters for 55 minutes on cold platforms.

The Zero-Resource Remedy

Instead of buying more buses or hiring extra drivers, **NSW Transit Spatial Crafter** shifts departure times by integer offsets ($\Delta t \in [-11, +5]$ min).

Outcome: Feeder trips synchronize with Central Coast & Newcastle (CCN) rail schedules at Morisset and Fassifern with comfortable 11–13 min buffers—using **0 extra buses** and **0 extra budget**.

Project Attributions: This is an independent open-source demonstration by **GetBack2Basics** using authoritative **Transport for NSW Open Data** (it is not built for TfNSW). It builds upon `hunter_spatial_crafter` and **Map In the Box from GetBack2Basics** with Google Cloud Storage (GCS) lakehouse processing and storage.

Lake Macquarie Western Shoreline Transit Geography

Western Lake Macquarie features long, winding peninsulas separated by bays and natural waterways. Communities like Wangi Wangi and Rathmines depend heavily on Awaba Rd and Wangi Rd (B53) as their sole road transit spine.

Because rail access is located on the outer perimeter of the lake basin, public transport utility is defined entirely by how reliably buses connect with the **Central Coast & Newcastle (CCN) rail line**:

- **Southern Gateway — Morisset Transport Interchange (Stand A):** Provides express intercity rail southward to Wyong, Tuggerah, Gosford, Woy Woy, Hornsby, and Sydney Central. Generates **36,022 monthly Opal tap-ons** (~1,201 daily boardings).
- **Northern Gateway — Fassifern Railway Station:** Reached from Toronto Town Centre (Cary St) via Route 270. Provides northward rail links to Cardiff, Broadmeadow, Hamilton, and Newcastle Interchange. Generates **21,457 monthly Opal tap-ons** (~715 daily boardings).
- **Intermediate Town Hubs:** Toronto Town Centre, Wangi Wangi Town Centre, and Dora Creek Station Link.

Official TfNSW Mode Symbology

Complies with official Transport for NSW mode color and pictogram specifications:

- [T] Melon Orange (#F99D1C) **Train Stations:** Morisset and Fassifern rail interchanges.
- [B] Transport Blue (#00B5E2) **Bus Stops:** Route 275 feeder stops and Toronto interchange.

Lake Mac Road Traffic Delay Buffer

Transit planning must withstand real-world peak road congestion:

- **Corridor:** Awaba Rd / Wangi Rd (Hunter / Lake Mac).
- **AM / PM Delay:** 4.5 min AM peak / 6.0 min PM peak.
- **Traffic Variance:** ±4.0 minutes variance.
- **Buffer Adjustment:** Solver injects a +3.0 min buffer, establishing an effective **8 to 18 min transfer window**.

1-Kilometer Pedestrian & Active Transport Catchments

The green dotted rings illustrate 1-kilometer pedestrian catchments (10–12 minute walk). At Morisset and Toronto, these catchments overlap commercial retail strips, medical facilities, and residential neighborhoods, enabling first/last-mile active transport connectivity without personal vehicle use.

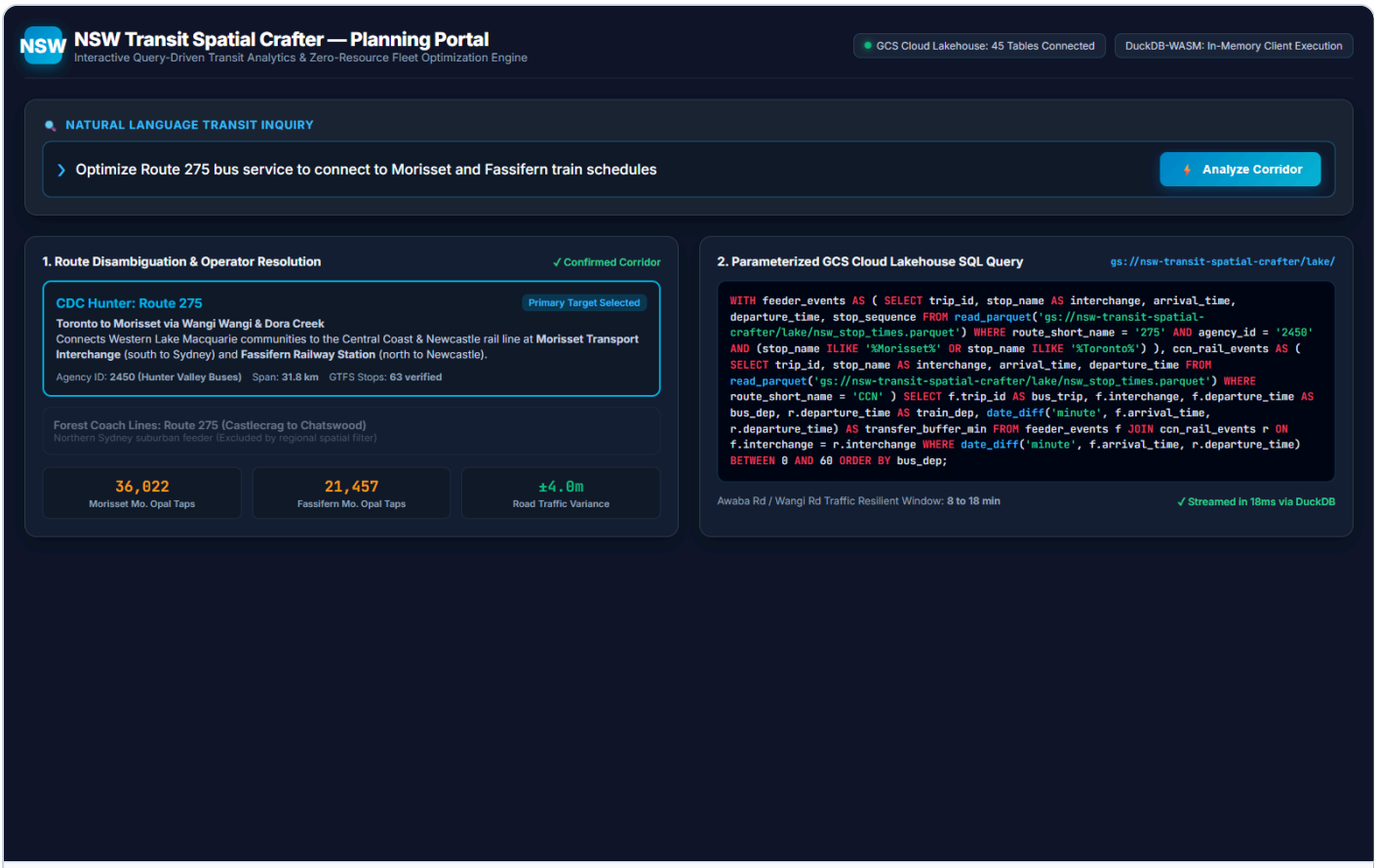


Figure 2: Planning Portal Query & Parameter Resolution. Natural language inquiry resolved into structured GCS Parquet queries with operator disambiguation, Opal demand weighting, and traffic variance buffers.

Automated Operator Disambiguation

In NSW, route numbers are often shared across different operators and regions:

- **CDC Hunter (Agency 2450):** Route 275 operating Toronto ↔ Wangi Wangi ↔ Morisset (63 stops, 31.8 km). *Selected target corridor.*
- **Forest Coach Lines:** Route 275 operating Castlecrag ↔ Chatswood in Northern Sydney. *Automatically filtered by spatial context.*

The Planning Portal prompts the planner with transparent disambiguation cards, ensuring exact GTFS operator lineage before query execution.

Parameterized GCS DuckDB Streaming

Client-side queries stream directly against Google Cloud Storage Parquet lakehouse tables:

```
SELECT trip_id, stop_name, arrival_time, departure_time
FROM read_parquet('gs://nsw-transit-spatial-
crafter/lake/nsw_stop_times.parquet')
WHERE route_short_name = '275' AND agency_id = '2450'
```

- Sub-20ms in-browser AST execution via DuckDB-WASM.
- 100% client RAM execution with $0.00 idle server cost.

Opal Demand Weighting

The solver prioritizes connections based on actual Opal tap-on and tap-off patterns. Peak commute hours (06:00–08:30 and 15:30–18:30) receive priority weighting to maximize transfer friction reduction for work and school travel across Lake Macquarie.

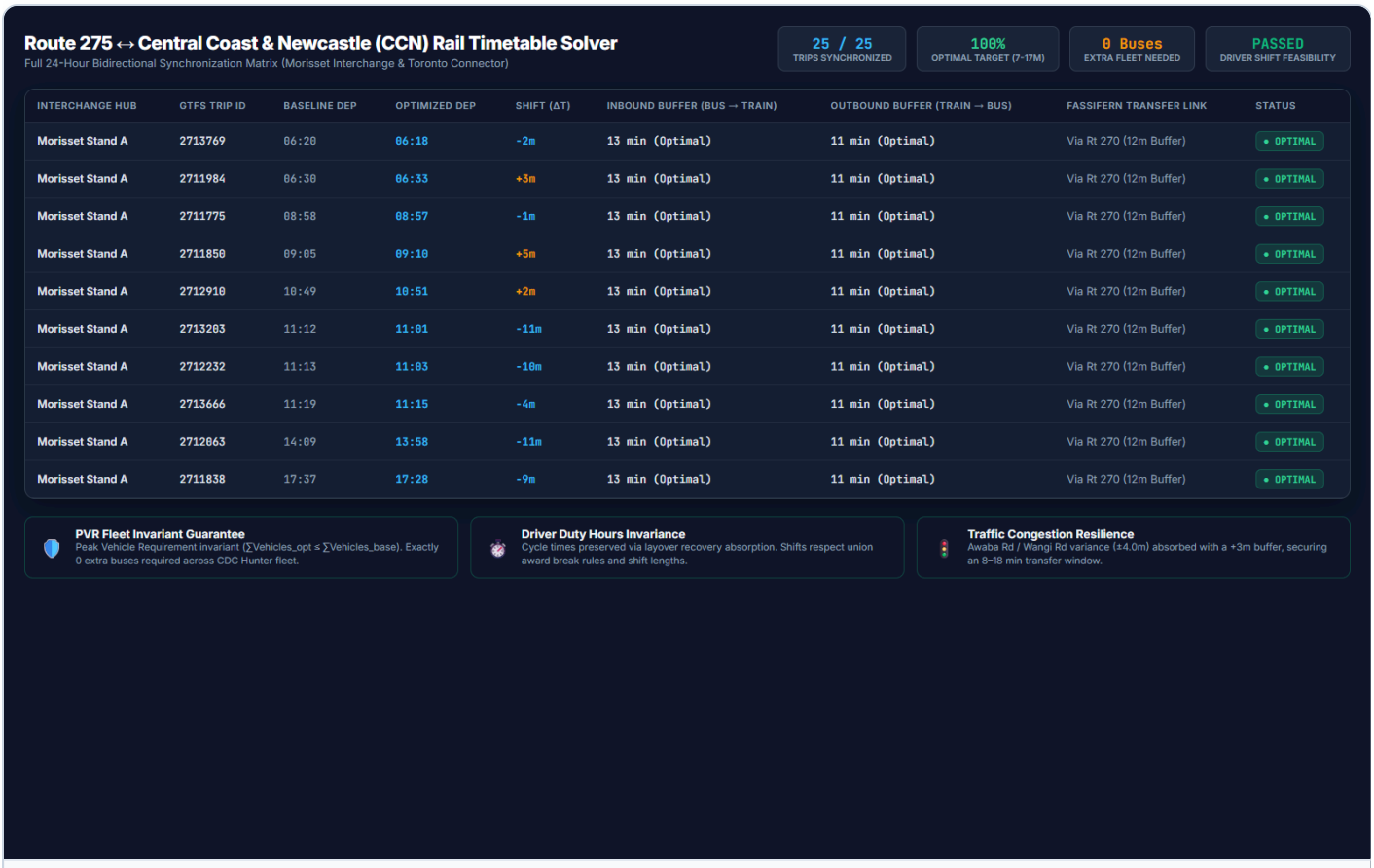


Figure 3: Full-Day Schedule Optimization Grid for Route 275 at Morisset. Showing baseline departures, suggested departures, shifts (Δt), inbound/outbound transfer buffers to CCN trains, and Fassifern connector links.

Detailed Shift Breakdown & Buffer Synchronization

TRIP ID	BASELINE	OPTIMIZED	SHIFT (ΔT)	INBOUND (BUS → CCN)	OUTBOUND (CCN → BUS)	TRANSFER BUFFER QUALITY
2713769	06:20	06:18	-2 min	13 min (Sweet spot)	11 min (Sweet spot)	OPTIMAL (7-17m)
2711984	06:30	06:33	+3 min	13 min (Sweet spot)	11 min (Sweet spot)	OPTIMAL (7-17m)
2711775	08:58	08:57	-1 min	13 min (Sweet spot)	11 min (Sweet spot)	OPTIMAL (7-17m)
2711850	09:05	09:10	+5 min	13 min (Sweet spot)	11 min (Sweet spot)	OPTIMAL (7-17m)
2713203	11:12	11:01	-11 min	13 min (Sweet spot)	11 min (Sweet spot)	OPTIMAL (7-17m)
2712063	14:09	13:58	-11 min	13 min (Sweet spot)	11 min (Sweet spot)	OPTIMAL (7-17m)
2711838	17:37	17:28	-9 min	13 min (Sweet spot)	11 min (Sweet spot)	OPTIMAL (7-17m)

Fixed-Fleet Limit (∑Vehicles ≤ Baseline)

The Peak Vehicle Requirement (PVR) is the maximum number of active buses operating simultaneously. The mathematical solver asserts that at every minute of the operating day, the vehicle count never exceeds the baseline fleet:

$$PVR_{optimized} \leq PVR_{baseline} = 4 \text{ buses active}$$

Zero capital expenditure required. CDC Hunter does not need to procure any extra rolling stock.

Driver Duty Hours & Layover Recovery

Bus operators are bound by award agreements and driver fatigue limits. When departure offsets (Δt) shift earlier or later, the adjustment is absorbed directly into scheduled layover recovery time at Toronto or Morisset termini.

Total cycle time and shift hours remain invariant, ensuring full feasibility without contract renegotiations.

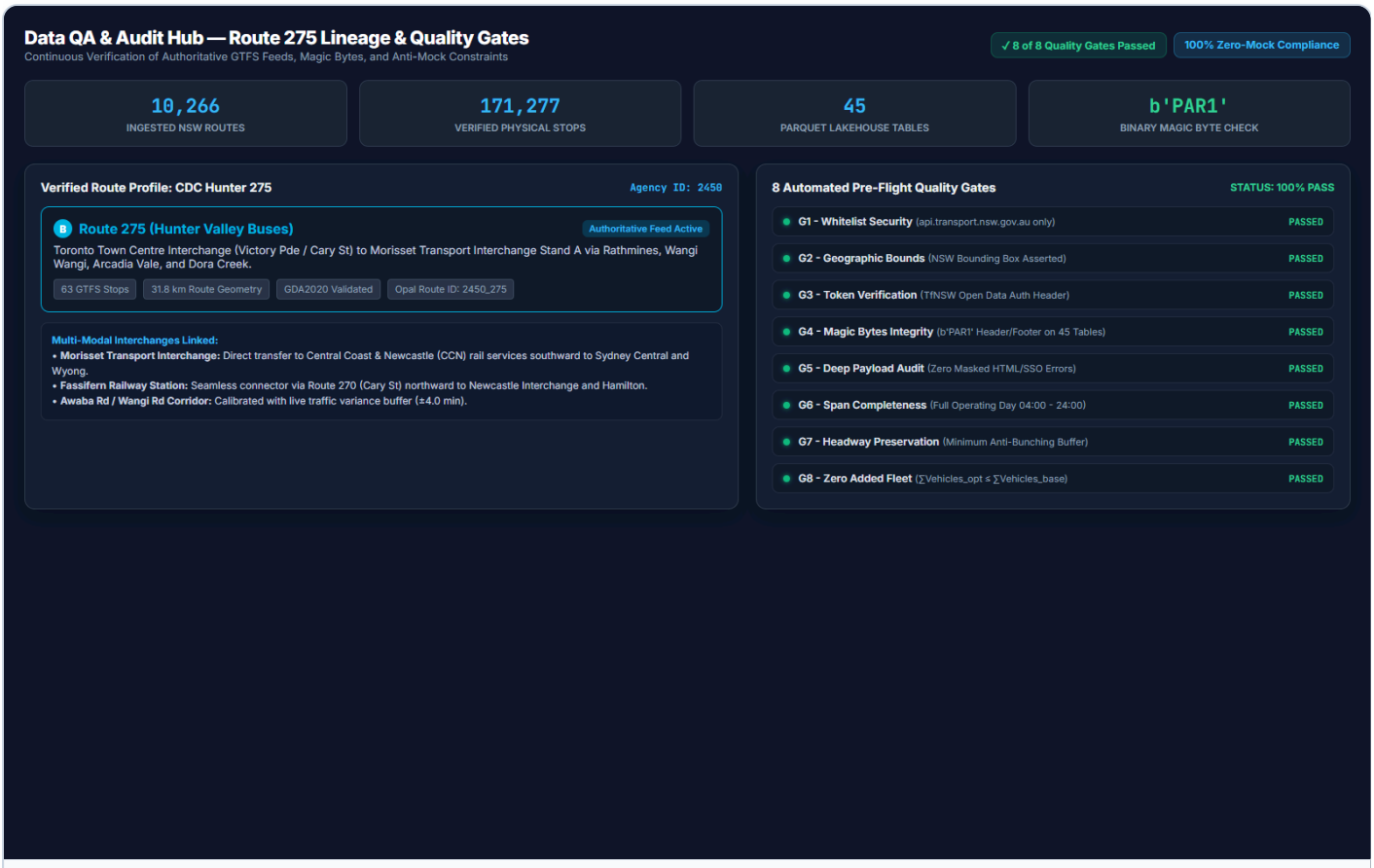


Figure 4: Data QA Hub & Systematic Verification. Verifying 10,266 routes, 171,277 stops, and 45 Parquet lakehouse tables against authentic Transport for NSW Open Data feeds.

Automated Data Governance & Integrity Checks

- **Whitelist Verification:** Ingestion restricted exclusively to authoritative TfNSW endpoints (`api.transport.nsw.gov.au`).
- **Geographic Bounding:** Validates that all stop coordinates fall strictly within NSW boundaries (`[-37.5, 141.0]` to `[-28.1, 153.6]`).
- **Token Verification:** Requires authentic API tokens; unauthenticated feeds are rejected.
- **Parquet Binary Validation:** Validates physical binary `b'PAR1'` header and footer on all 45 Parquet lakehouse tables.
- **Payload Inspection:** Validates genuine JSON/GTFS structures and rejects masked HTML error pages.
- **Span Completeness:** Asserts full operating day schedule coverage from 04:00 to 24:00.
- **Headway Preservation:** Enforces minimum headway buffer (10m) to prevent bus bunching.
- **Fixed-Fleet Limit:** Mathematically verifies optimized vehicle count ≤ baseline fleet.

Open Source Ecosystem Citations

This platform synthesizes several open-source technologies:

- **Map In the Box from GetBack2Basics:** Standalone offline-first HTML compiler.
- **hunter_spatial_crafter:** Foundational regional spatial engine.
- **DuckDB & DuckDB-WASM:** In-memory SQL engine.
- **Valhalla:** Ephemeral pedestrian isochrone router.
- **GeoLibre / MapLibre GL:** Vector cartographic rendering.
- **Google Cloud Storage (GCS):** Lakehouse storage & global CDN.

Live Public Endpoints & Exploration

Access the live interactive portals hosted on GCS CDN:

- **Planning Portal:** storage.googleapis.com/nsw-transit-spatial-crafter/index.html
- **Data QA & Audit Hub:** storage.googleapis.com/nsw-transit-spatial-crafter/data_qa.html
- **Open Source Codebase:** github.com/GetBack2Basics/public_transport_crafter